

Nicholas Ketz, PhD

Applied Research Scientist: Artificial Intelligence, Bioinformatics, Mathematical Models, Human Studies

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Summary

I am an applied scientist rooted in cognitive neuroscience. I have developed novel solutions to complex problems across multiple domains for 15+ years, with roles spanning the full AI lifecycle in both academic and industrial research labs.

Relevant Experience

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|---|--------------------------|
| Large Language Modeling | <i>03/2022 - Present</i> |
| • Developing and deploying DNA Large Language Models for cell and gene therapy optimization and cross-species engineering | |
| Generative AI | <i>01/2020 - 01/2022</i> |
| • Created AI systems for generating video and images grounded in learned human concepts and semantic understanding | |
| Model Based Reinforcement Learning | <i>01/2020 - 01/2022</i> |
| • Developed agent-based domain adaptation techniques including generative models supporting autonomous driving | |
| Lifelong Deep Learning | <i>09/2018 - 01/2021</i> |
| • Key personnel in DARPA's L2M program, developed methods to preserve performance across large sequential distribution shifts | |
| Closed-loop Neural Stimulation | <i>09/2016 - 06/2018</i> |
| • Developed devices and algorithms for closed-loop neural stimulation in human subjects to enhance cognitive performance | |

Education/Employment

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|--|--------------------------|
| Principal AI/ML Scientist <i>Colossal Biosciences/Form Bio</i> | <i>02/2022 - Present</i> |
| • Advise and review technical direction for a cross-disciplinary science team; lead development directly on high-stakes projects | |
| Research Scientist: Information and Systems Sciences <i>HRL Laboratories</i> | <i>09/2016 - 01/2022</i> |
| • Developed novel AI methods funded by Department of Defense with product development for aerospace and automotive industries | |
| PhD: Computational Cognitive Neuroscience <i>University of Colorado, Boulder</i> | <i>09/2010 - 09/2016</i> |
| • Developed biologically inspired neural networks informed by neuroimaging experiments of cognitive function | |
| Research Assistant: Davachi Human Memory Lab <i>New York University</i> | <i>09/2007 - 07/2010</i> |
| • Supported NIH R01 research for neuroimaging of the human long-term memory system | |
| BA: Physics, minor: Psychology <i>University of Minnesota, Twin Cities</i> | <i>09/2003 - 06/2007</i> |
| • Studied physics with a minor in psychology, with research experience in particle physics and cognitive neuroscience | |

Skills

- **Technical:** Deep Learning (CNN, RNN, Transformers, RL), Machine Learning (Optimization, Unsupervised Learning), Statistics (Bayesian, Parametric/Non-parametric, Time-series), Programming (Python, PyTorch, TensorFlow, scikit-learn, Docker, Git, Bash, SQL), Agentic AI (LLM agents, tool-use orchestration, agentic coding workflows), Cloud (GCP)
- **AI Lifecycle:** Full-cycle AI development (proof-of-concept to production), cross-team product development, agentic workflows for code and product delivery, containerized system design, CI/CD pipelines, cross-container unit testing
- **Professional:** Team leadership, cross-departmental collaboration (engineering, product, sales), client/stakeholder management, project coordination across multiple research domains and organizations

Select Publications/Patents/White Papers

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|---|-------------------------------|
| System and method for semantically grounded video generation | <i>US Patent, 2024</i> |
| • Generative video AI system prompted by learned human concepts | |
| A domain-agnostic approach for characterization of lifelong learning systems | <i>Neural Networks, 2023</i> |
| • Metrics framework and theory for continual learning in deep neural network systems across multiple domains and tasks | |
| Model Interpretability Methods to Predict Gene Therapy Manufacturing Failures | <i>White Paper, 2022</i> |
| • Extracting learned motifs from DNA sequence models predictive of gene therapy manufacturing failures | |
| Sliced Cramér synaptic consolidation for preserving deeply learned representations | <i>ICLR, 2020 (Spotlight)</i> |
| • Framework for overcoming catastrophic forgetting by preserving the distribution of the network's output at an arbitrary layer | |